

HKCEE 1991

Mathematics II

91 $(a^{2a})(3a^{4a})$

1.

- A. $3a^{6a}$
- B. $(3a)^{6a}$
- C. $3a^{8a}$
- D. $4a^{6a}$
- E. $(3^{4a})(a^{6a})$

91 $\frac{1}{1-x^2} - \frac{1}{(1+x)^2} =$

2.

- A. $\frac{2}{(1-x^2)(1+x^2)}$
- B. $\frac{2x^2}{(1-x^2)(1+x^2)}$
- C. $\frac{2x^2}{(1-x^2)(1+x)^2}$
- D. $\frac{2}{(1-x)(1+x)^2}$
- E. $\frac{2x}{(1-x)(1+x)^2}$

91 Which one of the following is a factor
3. of $x^3 - 4x^2 + x + 6$?

- A. $(x+1)(x-2)$
- B. $(x+1)(x+2)$
- C. $(x-1)(x+2)$
- D. $(x-1)(x-3)$
- E. $(x-1)(x+3)$

91 If $y = \sqrt{\frac{1+mx}{1-mx}}$, then $x =$

4.

- A. $\frac{m(y-1)}{y+1}$
- B. $\frac{y-1}{m(y+1)}$

C. $\frac{(1-y^2)}{m(1+y^2)}$

D. $\frac{m(y^2-1)}{(y^2+1)}$

E. $\frac{(y^2-1)}{m(y^2+1)}$

91 $\frac{1}{x^2} + \frac{1}{y^2} =$
5. $\frac{\frac{1}{x} + \frac{1}{y}}{\frac{1}{x} + \frac{1}{y}} =$

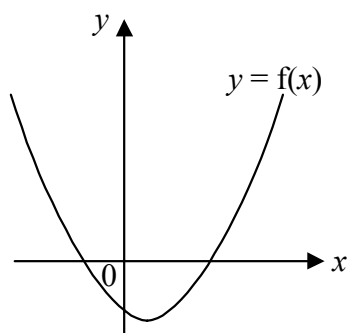
- A. $\frac{1}{x^2} + \frac{1}{y^2}$
- B. $\frac{1}{x^2} + \frac{1}{xy} + \frac{1}{y^2}$
- C. $\frac{1}{x^2} + \frac{2}{xy} + \frac{1}{y^2}$
- D. $\frac{1}{x^2} - \frac{2}{xy} + \frac{1}{y^2}$
- E. $\frac{1}{x^2} - \frac{1}{xy} + \frac{1}{y^2}$

91 The L.C.M. of x , $2x^2$, $3x^3$, $4x^4$, $5x^5$ is
6.

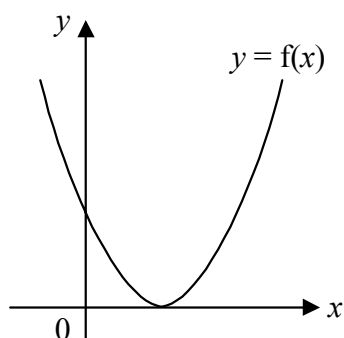
- A. x
- B. $5x^5$
- C. $60x^5$
- D. $120x^5$
- E. $120x^{15}$

91 In which of the following cases the
7. equation $f(x) = 0$ **cannot** be solved by the method of bisection?

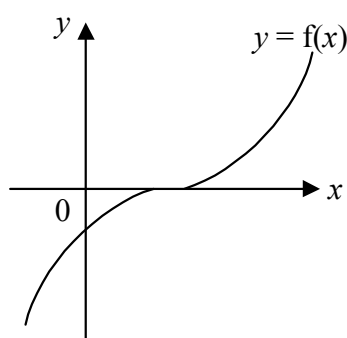
A.



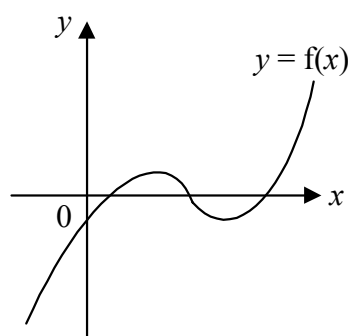
B.



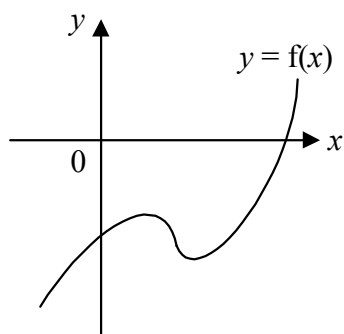
C.



D.



E.



91 Solve the following equations :

8. $x - 1 = y + 2 = x + y - 5$

A. $x = 1, y = -2$

B. $x = 1, y = 4$

C. $x = 4, y = 1$

D. $x = 7, y = -2$

E. $x = 7, y = 4$

91

9.

Let y vary partly as $\frac{1}{x}$ and partly as x .

When $x = 1, y = 5$ and when $x = 4,$

$y = \frac{25}{2}$. Find y when $x = 2$.

A. $\frac{5}{2}$

B. 4

C. $\frac{25}{4}$

D. 7

E. $\frac{17}{2}$

91

10.

If $\frac{1}{a} : \frac{1}{b} = 2 : 3$ and $a : c = 4 : 1$, then

$a : b : c =$

A. $12 : 8 : 3$.

B. $8 : 3 : 2$.

C. $4 : 6 : 1$.

D. $2 : 3 : 8$.

E. $2 : 3 : 4$.

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11.

A blanket loses 10% of its length and 8% of its width after washing. The percentage loss in area is

A. 18.8% .

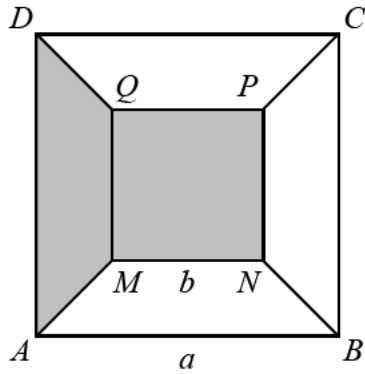
B. 18% .

C. 17.2% .

D. 9% .

E. 8% .

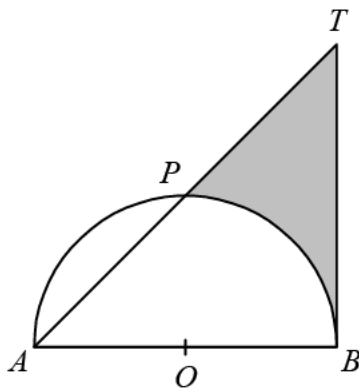
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12.



In the figure, $ABCD$ is a square of side a and $MNPQ$ is a square of side b . The four trapeziums are identical. The area of the shaded region is

- A. $\frac{3b^2 + a^2}{4}$
- B. $\frac{3b^2 - a^2}{2}$
- C. $\frac{5b^2 + a^2}{4}$
- D. $\frac{5b^2 - a^2}{4}$
- E. $\frac{(a-b)^2}{4} + b^2$

91
13.

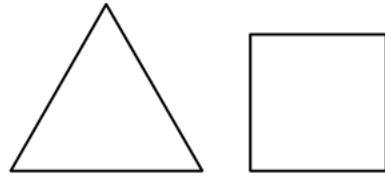


In the figure, TB touches the semi-circle at B . TA cuts the semi-circle at P such that $TP = PA$. If the radius of the semi-circle is 2, find the area of the shaded region.

- A. $12 - \pi$
- B. $8 - \pi$

- C. $6 - \pi$
- D. $4 - \pi$
- E. $2(4 - \pi)$

91
14.



An equilateral triangle and a square have equal perimeters.

$$\frac{\text{Area of the triangle}}{\text{Area of the square}} =$$

- A. $\frac{9\sqrt{3}}{16}$
- B. $\frac{\sqrt{3}}{4}$
- C. $\frac{\sqrt{3}}{3}$
- D. $\frac{4\sqrt{3}}{9}$
- E. 1

91
15. A man borrows \$10 000 from a bank at 12% per annum compounded monthly. He repays the bank \$2000 at the end of each month. How much does he still owe the bank just after the second repayment?

- A. \$6181
- B. \$6200
- C. \$6201
- D. \$8304
- E. \$8400

91
16. $\left[\frac{1}{\cos \theta} + \tan \theta \right] (1 - \sin \theta) =$

- A. $\sin \theta$
- B. $\cos \theta$
- C. $\cos^2 \theta$
- D. $1 + \sin \theta$
- E. $\sin \theta \tan \theta$

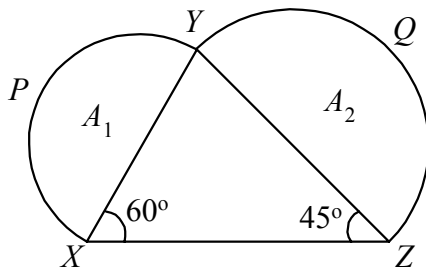
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17. $\frac{\sin(\theta - 90^\circ)}{\tan(\theta + 180^\circ)} =$

- A. $\cos \theta$
 B. $-\cos \theta$
 C. $\frac{\cos^2 \theta}{\sin \theta}$
 D. $-\frac{\cos^2 \theta}{\sin \theta}$
 E. $\frac{1}{\sin \theta}$

91
18. For $0 \leq \theta < 2\pi$, how many roots does the equation $\tan \theta + 2 \sin \theta = 0$ have?

- A. 1
 B. 2
 C. 3
 D. 4
 E. 5

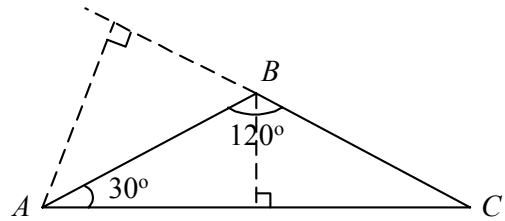
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19.



In the figure, XPY and YQZ are semi-circles with areas A_1 and A_2 respectively. $\angle YXZ = 60^\circ$ and $\angle YZX = 45^\circ$. The ratio $A_1 : A_2 =$

- A. $\sqrt{2} : \sqrt{3}$
 B. $\sqrt{2} : 3$
 C. $2 : 3$
 D. $2 : \sqrt{3}$
 E. $\sqrt{3} : \sqrt{2}$

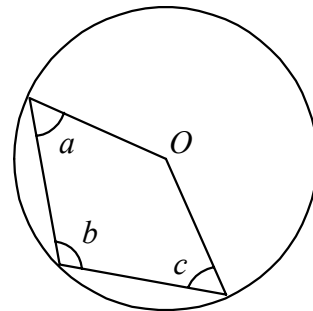
91
20.



In the figure, $\angle A = 30^\circ$ and $\angle B = 120^\circ$. The ratio of the altitudes of the triangle ABC from A and from B is

- A. $2 : 1$
 B. $\sqrt{3} : 1$
 C. $\sqrt{2} : 1$
 D. $1 : \sqrt{2}$
 E. $1 : \sqrt{3}$

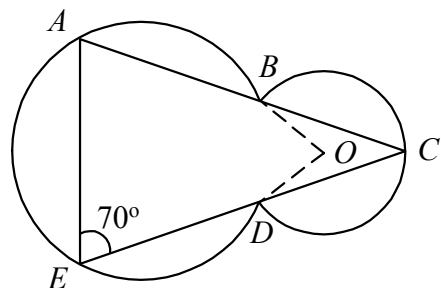
91
21.



In the figure, O is the centre of the circle. Find $a + c$.

- A. b
 B. $2b$
 C. $180^\circ - b$
 D. $360^\circ - b$
 E. $360^\circ - 2b$

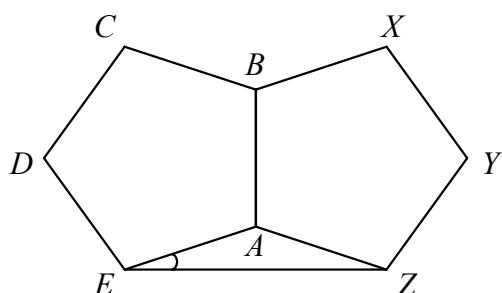
91
22.



In the figure, O is the centre of the circle BCD . ABC and EDC are straight lines. $BC = DC$ and $\angle AED = 70^\circ$. Find $\angle BOD$.

- A. 40°
- B. 70°
- C. 80°
- D. 90°
- E. 140°

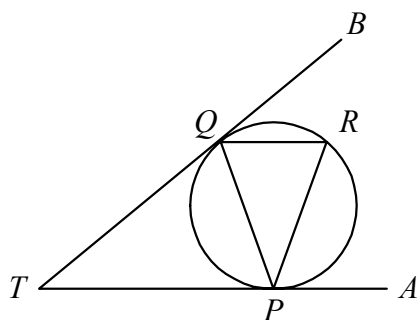
91
23.



In the figure, $ABCDE$ and $ABXYZ$ are two identical regular pentagons. Find $\angle AEZ$.

- A. 15°
- B. 18°
- C. 24°
- D. 30°
- E. 36°

91
24.

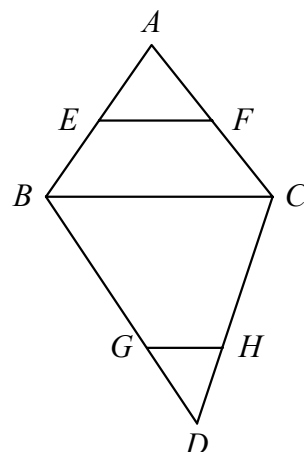


In the figure, TPA and TQB are tangents to the circle at P and Q respectively. If $PQ = PR$, which of the following **must** be true?

- I. $\angle APR = \angle QRP$
- II. $\angle QTP = \angle QPR$
- III. $\angle QPR = \angle APR$

- A. I only
- B. II only
- C. III only
- D. I and II only
- E. I and III only

91
25.



In the figure, E and F are the mid-points of AB and AC respectively. G and H divide DB and DC respectively in the ratio $1 : 3$. If $EF = 12$, find GH .

- A. 3
- B. 4
- C. 6
- D. 8
- E. 12

91 The circle $x^2 + y^2 + 4x + ky + 4 = 0$
26. passes through the point $(1, 3)$. The radius of the circle is

- A. $\sqrt{68}$
- B. $\sqrt{48}$
- C. $\sqrt{17}$
- D. 6
- E. 3

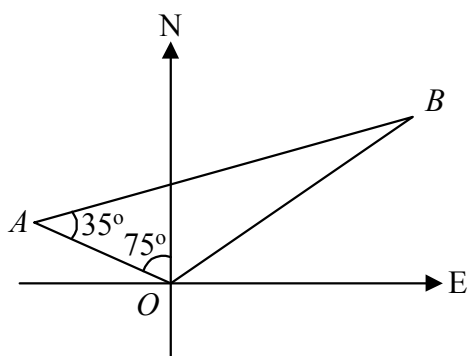
91 Let A and B be the points $(4, -7)$ and
27. $(-6, 5)$ respectively. The equation of the line passing through the mid-point of AB and perpendicular to $3x - 4y + 14 = 0$ is

- A. $3x - 4y - 1 = 0$.
 B. $3x + 4y + 7 = 0$.
 C. $4x - 3y + 1 = 0$.
 D. $4x + 3y - 7 = 0$.
 E. $4x + 3y + 7 = 0$.

91 $PQRS$ is a parallelogram with vertices $P = (0, 0)$, $Q = (a, b)$ and $S = (-b, a)$. Find R .

- A. $(-a, -b)$
 B. $(a, -b)$
 C. $(a - b, a - b)$
 D. $(a - b, a + b)$
 E. $(a + b, a + b)$

91
29.



In the figure, A and B are the positions of two boats. The bearing of B from A is

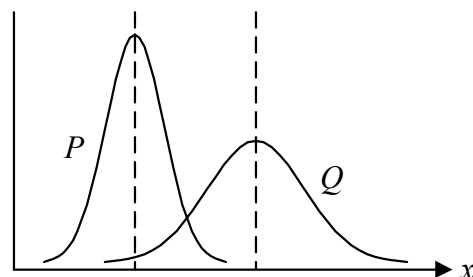
- A. $N55^\circ E$.
 B. $N70^\circ E$.
 C. $N20^\circ E$.
 D. $S35^\circ E$.
 E. $S75^\circ E$.

91 The mean and standard deviation of a distribution of test scores are m and s respectively. If 4 marks are added to each score of the distribution, what are the mean and standard deviation of the new distribution?

- | | Mean | Standard Deviation |
|----|---------|--------------------|
| A. | $m + 4$ | s |
| B. | $m + 4$ | $s + 2$ |
| C. | $m + 4$ | $s + 4$ |

- D. m $s + 2$
 E. m $s + 4$

91
31.



The graph shows the frequency curves of two symmetric distributions P and Q . Which of the following is /are true?

- I. The mean of $P <$ the mean of Q .
 II. The mode of $P >$ the mode of Q .
 III. The inter-quartile range of $P <$ the inter-quartile range of Q .

- A. I only
 B. I and II only
 C. I and III only
 D. II and III only
 E. I, II and III

91 A fair die is thrown 3 times. The probability that "6" occurs exactly once is

- A. $\frac{1}{3}$.
 B. $\left(\frac{1}{6}\right)^3$.
 C. $\frac{1}{3} \times \frac{1}{6}$.
 D. $\left(\frac{1}{6}\right)\left(\frac{5}{6}\right)^2$.
 E. $3\left(\frac{1}{6}\right)\left(\frac{5}{6}\right)^2$.

91 If $(\sqrt{3} + 1)\sqrt{x} = 2$, then $x =$
33.

- A. $2 - \sqrt{3}$.

- B. $\sqrt{3} - 1$.
 C. 1 .
 D. $2(2 - \sqrt{3})$.
 E. $4 - \sqrt{3}$.

91 If $\log x : \log y = m : n$, then $x =$
 34.

- A. $\frac{my}{n}$.
 B. $(m - n)y$.
 C. $m - n + y$.
 D. $\frac{m}{y^n}$.
 E. $\frac{m \log y}{n}$.

91 If $f(x) = x - \frac{1}{x}$, then $f(x) - f\left(\frac{1}{x}\right) =$
 35.

- A. 0 .
 B. $2x$.
 C. $-\frac{2}{x}$.
 D. $2\left(x - \frac{1}{x}\right)$.
 E. $2\left(\frac{1}{x} - x\right)$.

91 If $p(x^2 - x) + q(x^2 + x) = 4x^2 + 8x$, find
 36. p and q .

- A. $p = 4, q = 8$
 B. $p = -8, q = 4$
 C. $p = -2, q = 6$
 D. $p = 2, q = 6$
 E. $p = 6, q = -2$

91 If $x < 0 < y$, then which one of the
 37. following **must** be positive?

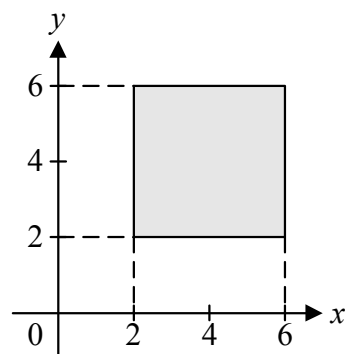
- A. $x + y$
 B. $x - y$
 C. $y - x$
 D. xy

E. $\frac{y}{x}$

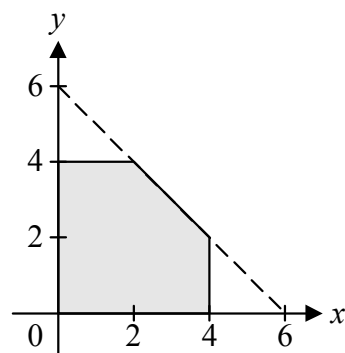
91 Which one of the following shaded
 38. regions represents the solution of

$$\begin{cases} 2 \leq x + y \leq 6 \\ 0 \leq x \leq 4 \\ 0 \leq y \leq 4 \end{cases} ?$$

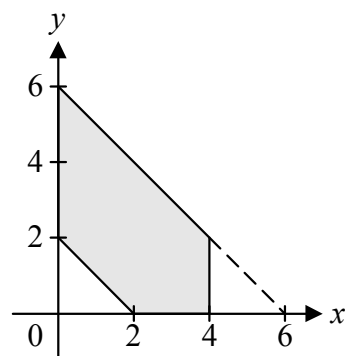
A.



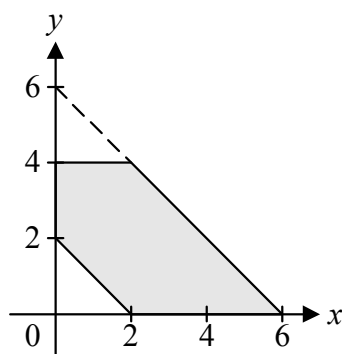
B.



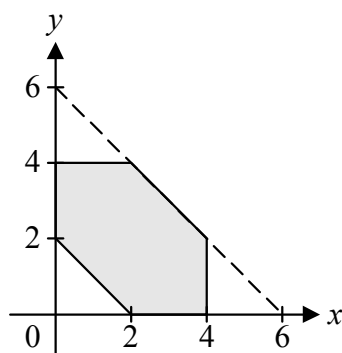
C.



D.



E.



- 91 If $(x - 2)(x - 3) = (a - 2)(a - 3)$, solve for x .

- A. $x = 0$ or 5
- B. $x = 2$ or 3
- C. $x = a$ or 2
- D. $x = a$ or 3
- E. $x = a$ or $5 - a$

- 91 If the sum to n terms of an A.P. is $n^2 + 3n$, find the 7th term of the A.P.

- A. 16
- B. 18
- C. 54
- D. 70
- E. It cannot be found.

- 91 If x, y, z are in G.P, which of the following **must** be true?

- I. $x + 3, y + 3, z + 3$ are in G.P.
 - II. $3x, 3y, 3z$ are in G.P.
 - III. x^2, y^2, z^2 are in G.P.
- A. I only
 - B. II only
 - C. III only

- D. I and II only
- E. II and III only

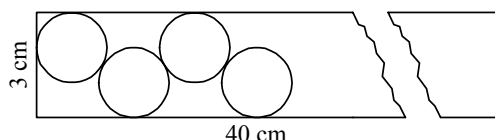
- 91 3 kg of a solution contains 40% of alcohol by weight. How much alcohol should be added to obtain a solution containing 50% of alcohol by weight?

- A. 0.3 kg
- B. 0.6 kg
- C. 0.75 kg
- D. 1.5 kg
- E. 3.75 kg

- 91 P sold an article to Q at a profit of 25%. Q sold it to R also at a profit of 25%. If Q gained \$500, how much did P gain?

- A. \$250
- B. \$320
- C. \$333
- D. \$400
- E. \$500

- 91
44.



From a rectangular metal sheet of width 3 cm and length 40 cm, at most how many circles each of radius 1 cm can be cut?

- A. 20
- B. 21
- C. 22
- D. 23
- E. 24

DIRECTIONS: Question 45 and 46 refer to the figure below, which shows a cuboid $ABCDEFGH$ with $AE = 2a$, $EF = 2b$ and $FG = 2c$. AC and BD intersect at X .

A. $\sqrt{a^2 + b^2 + c^2}$
 B. $\sqrt{a^2 + b^2 + (2c)^2}$
 C. $\sqrt{a^2 + (2b)^2 + c^2}$
 D. $\sqrt{(2a)^2 + b^2 + c^2}$
 E. $2\sqrt{a^2 + b^2 + c^2}$

A. $\frac{a}{b}$.

B. $\frac{2a}{b}$.

C. $\frac{\sqrt{(2a)^2 + c^2}}{b}$.

D. $\frac{a}{\sqrt{b^2 + c^2}}$.

E. $\frac{2a}{\sqrt{b^2 + c^2}}$.

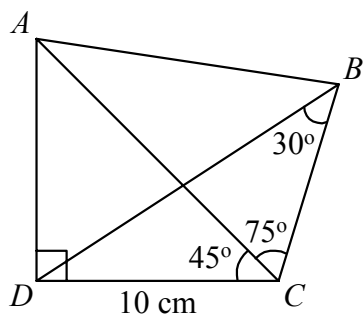
A. 0
B. 1
C. -1
D. 10
E. -10

A. $y = 2 \cos x$.
B. $y = 2 - \sin x$.
C. $y = 2 + \sin x$.
D. $y = 2 - \cos x$.
E. $y = 2 + \cos x$.

The diagram shows a 3D coordinate system with axes labeled South, East, and a vertical axis. A point P is located in the space. A line segment of length 30 m connects the origin O to a point on the South axis. The angle between the South axis and the line segment PO is 30° . The angle between the East axis and the line segment PO is 45° .

A. 7.5 m .
B. 15 m .
C. $15\sqrt{2}$ m .
D. $15\sqrt{3}$ m .
E. 45 m .

91
50.



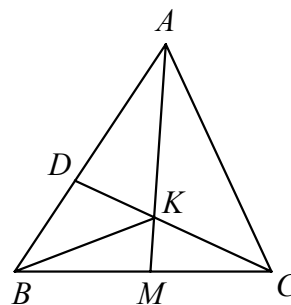
In the figure, arc AB : arc BC : arc CD : arc DE : arc EA = 1 : 2 : 3 : 4 : 5. Find θ .

- A. 30°
- B. 36°
- C. 60°
- D. 72°
- E. 120°

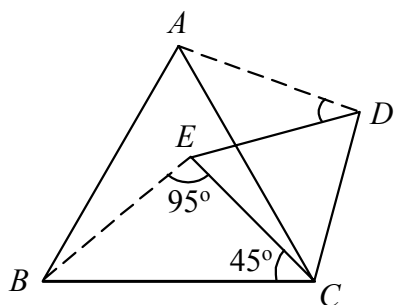
In the figure, find the length of AB , correct to the nearest cm.

91
53.

- A. 14 cm
- B. 15 cm
- C. 16 cm
- D. 17 cm
- E. 18 cm



91
51.



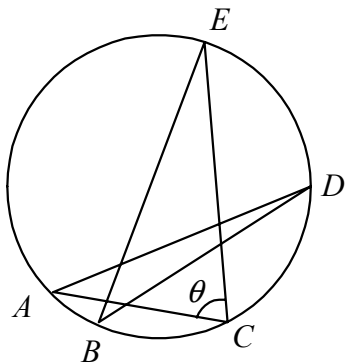
In the figure, M is the mid-point of BC and $AD = 2DB$. AM and CD intersect at K . Find $\frac{\text{area of } \triangle ADK}{\text{area of } \triangle AKC}$.

- A. $\frac{1}{2}$
- B. $\frac{2}{3}$
- C. $\frac{3}{4}$
- D. $\frac{4}{5}$
- E. 1

In the figure, ABC and CDE are equilateral triangles. Find $\angle ADE$.

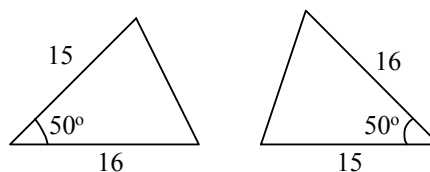
- A. 15°
- B. 35°
- C. 40°
- D. 45°
- E. 50°

91
52.

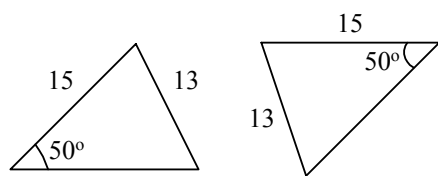


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54. In the figure, which of the pairs of triangles **must** be congruent?

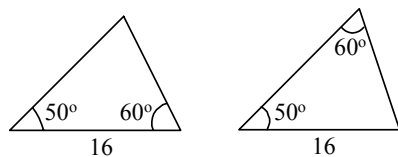
I.



II.



III.



- A. I only
- B. II only
- C. I and III only
- D. II and III only
- E. I, II and III